What's new

Visualizing the Gilbreath expectation sequence

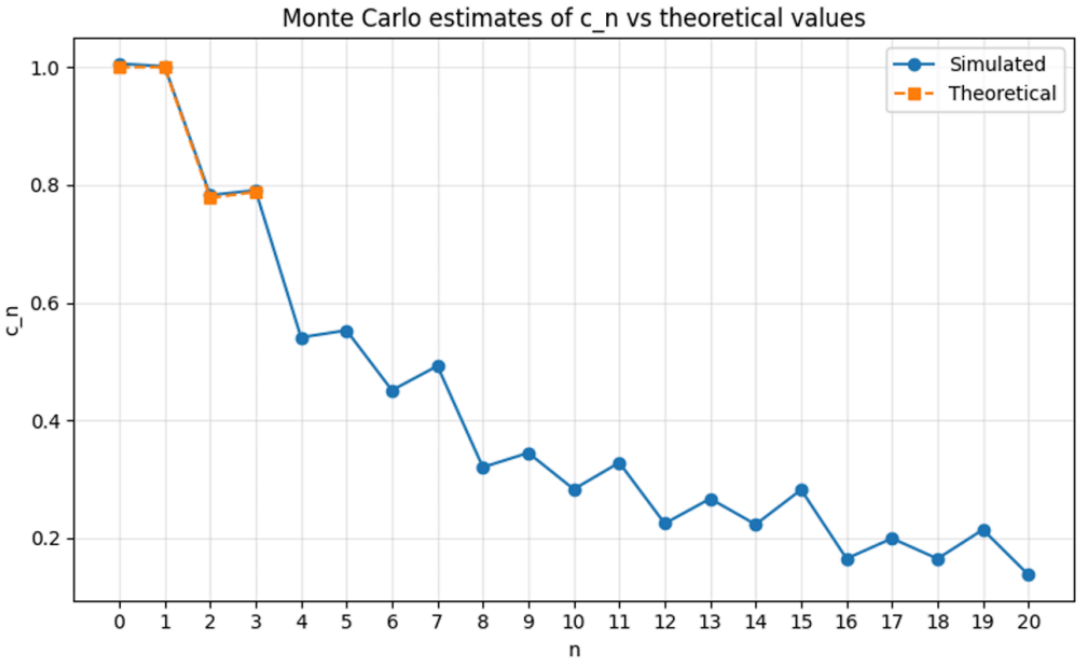
Terence Tao · 14 July 2026

https://terrytao.wordpress.com/2026/07/14/visualizing-the-gilbreath-expectation-sequence/

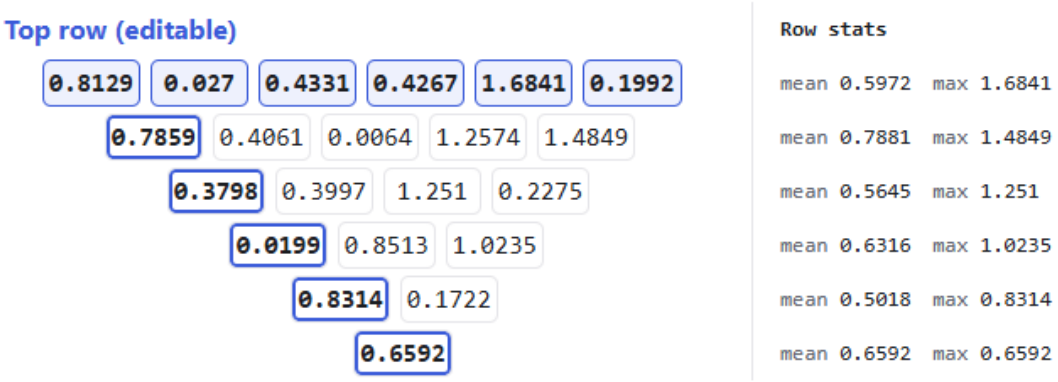
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One byproduct of learning how to use coding agents to create visualization apps is that it now becomes straightforward to convert any figure in one's papers that had already been generated by code (e.g., in Python) into a more interactive, animated applet.

I can illustrate this with Figure 1 from my recent paper on the Gilbreath conjecture with Chase and Hunter, reproduced below:



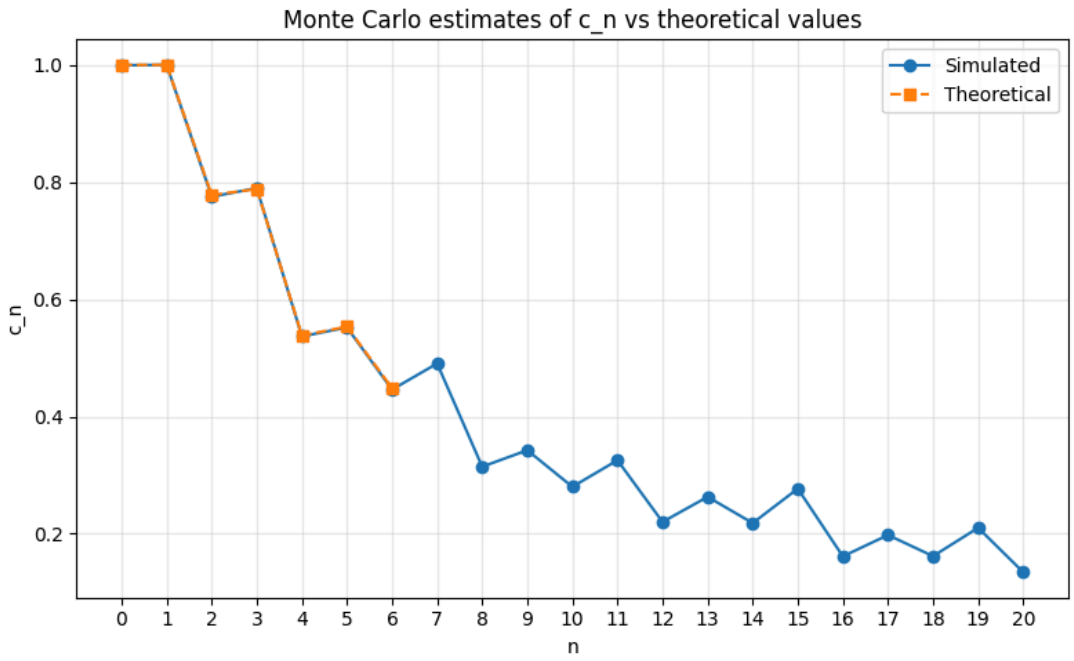
This plot displays both exact and numerically simulated values of a certain poorly understood sequence c_n relating to the Gilbreath conjecture, which I will call the "Gilbreath expectation sequence" here for lack of a better name. The definition of the sequence is as follows. Consider a "Gilbreath array" which is an inverted pyramid, where the top entries are $n+1$ independent exponential random variables of mean 1, and all the other entries are the absolute values of the differences of the two entries immediately above it. Thanks to the visualizer app, I can quickly give an example (with $n+1=6$):



The left diagonal entries are then random variables; the sequence $c_0, \dots, c_n$ are defined to be the expectation of these values. (The process is stationary, so in fact any entry on the ith row will have expectation c_i.)

If one starts with the first n normalized prime gaps (which have expectation about $\log n/2$, and are conjecturally distributed asymptotically according to a geometric distribution), then standard conjectures (e.g., the prime tuples conjecture) predict that the kth row entries should decay like $c_k \log n/2$, at least for small k. So the Gilbreath conjecture appears to be tied to how fast the sequence c_k decays with k.

One can in principle work out each value of c_k as an explicit rational number by performing a certain complicated multivariate integral, but in the paper we only did this for $k \leq 3$ (the orange line in the above figure); for the remaining k we performed a Monte Carlo simulation with 10^6 Gilbreath arrays to obtain a numerical approximation (in blue), which (as per the law of large numbers) agreed well with the theoretical values. A later calculation of Michael Ross extended the theoretical values to $k \leq 6$, maintaining the good fit:

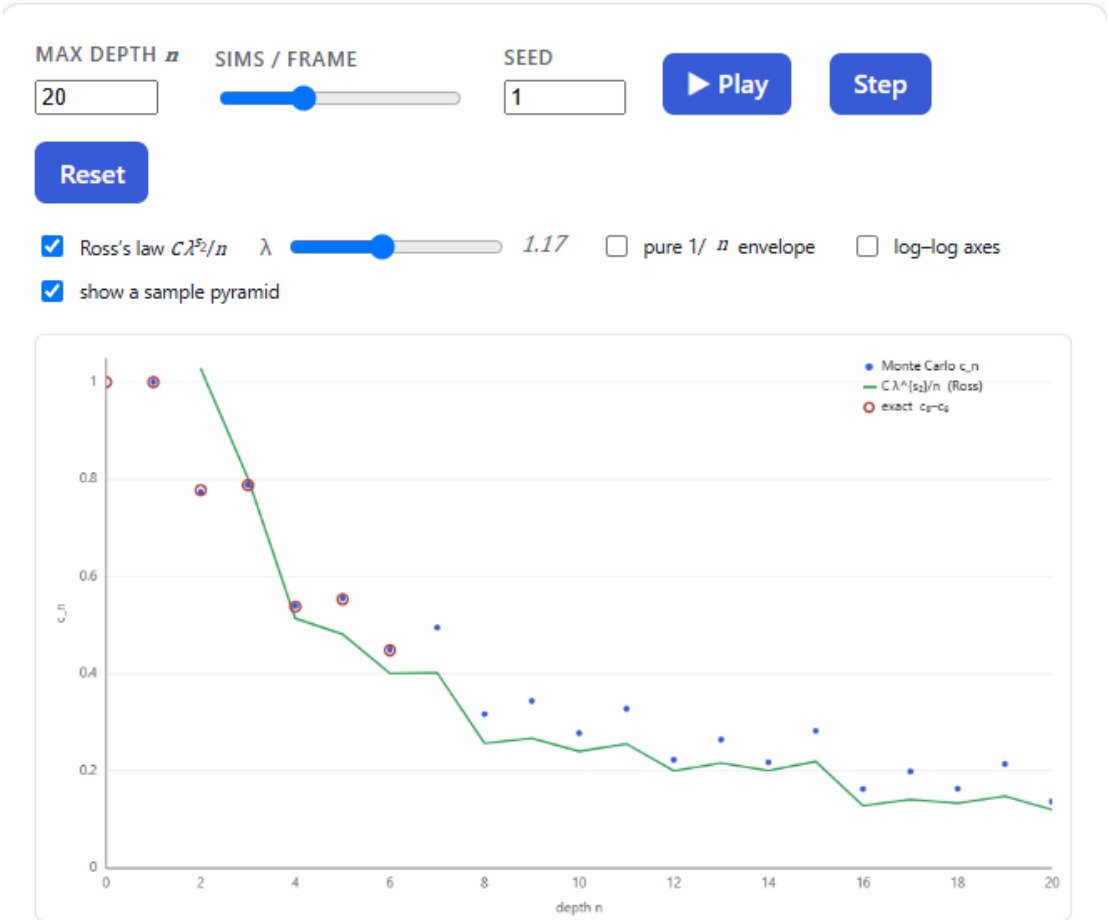


The asymptotic behavior of the sequence c_n remains mysterious. Clearly, it is not monotonic; in fact we cannot even prove it is bounded. The best we could do in our paper was establish an inequality which, roughly speaking, showed that c_n cannot decay faster than $1/n$.

In a recent preprint of Ross, these numerics were extended, and a rough empirical prediction

$$c_n \approx C\lambda^{s_2(n)}/n$$

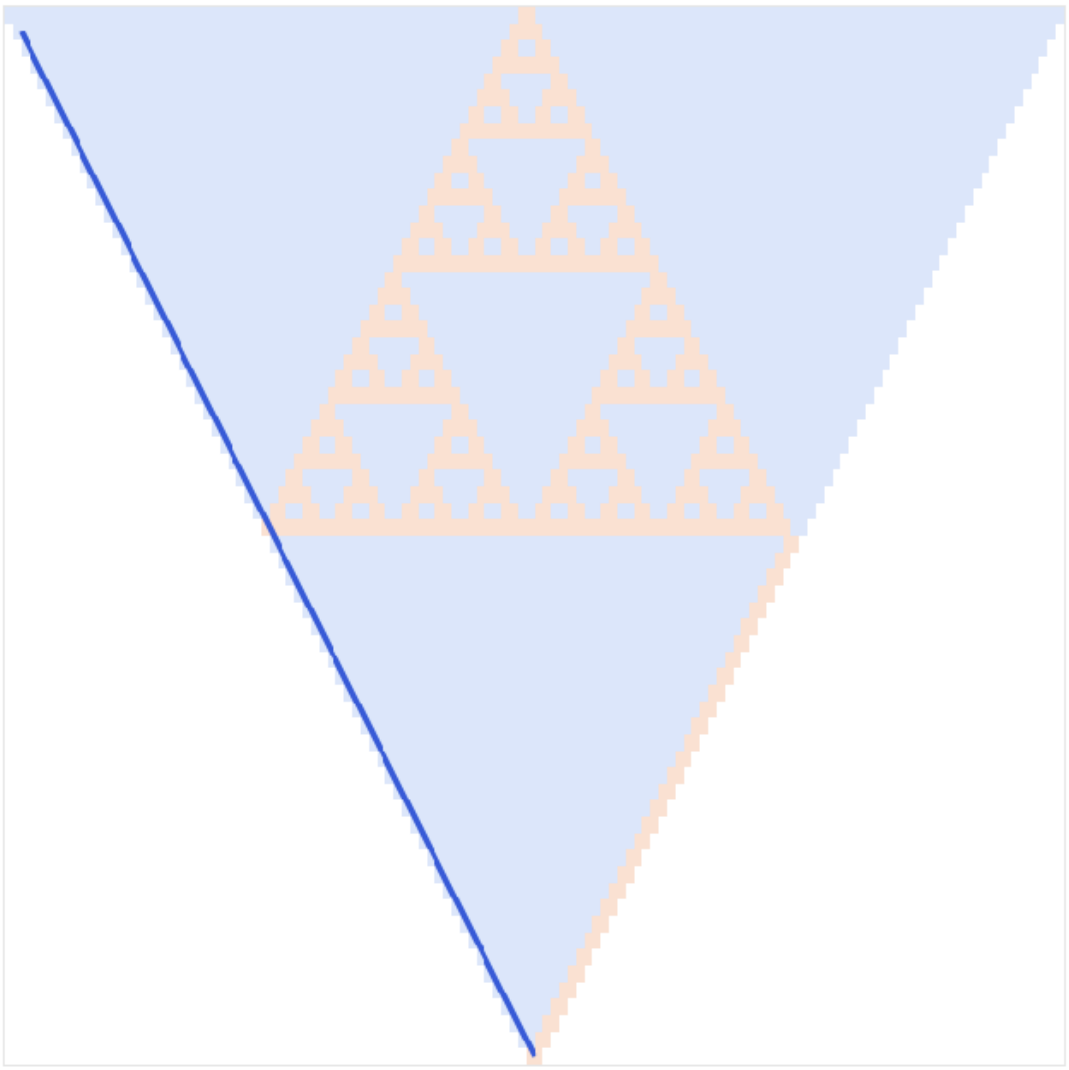
was proposed for some constants $C > 0$ and $\lambda > 1$ (empirically $\lambda \approx 1.17$), where $s_2(n)$ is the number of 1's in the binary expansion of n; in particular, it is the fluctuation in this quantity s_2 that is intended to explain much of the non-monotonic behavior of c_n. These are now all displayed in the following companion applet, which was a routine matter to generate in about an hour by the coding agent (which by this point has extensive experience with creating such apps, encoded via a "skill" markdown file that it maintains):



The appearance of the quantity $s_2(n)$ may initially appear mysterious, but it is related to Lucas's theorem, Kummer's theorem, and the Sierpinski gasket. Consider for instance a Gilbreath array where all the entries are zero except for a single "spike". Then the following Sierpinski pattern emerges:

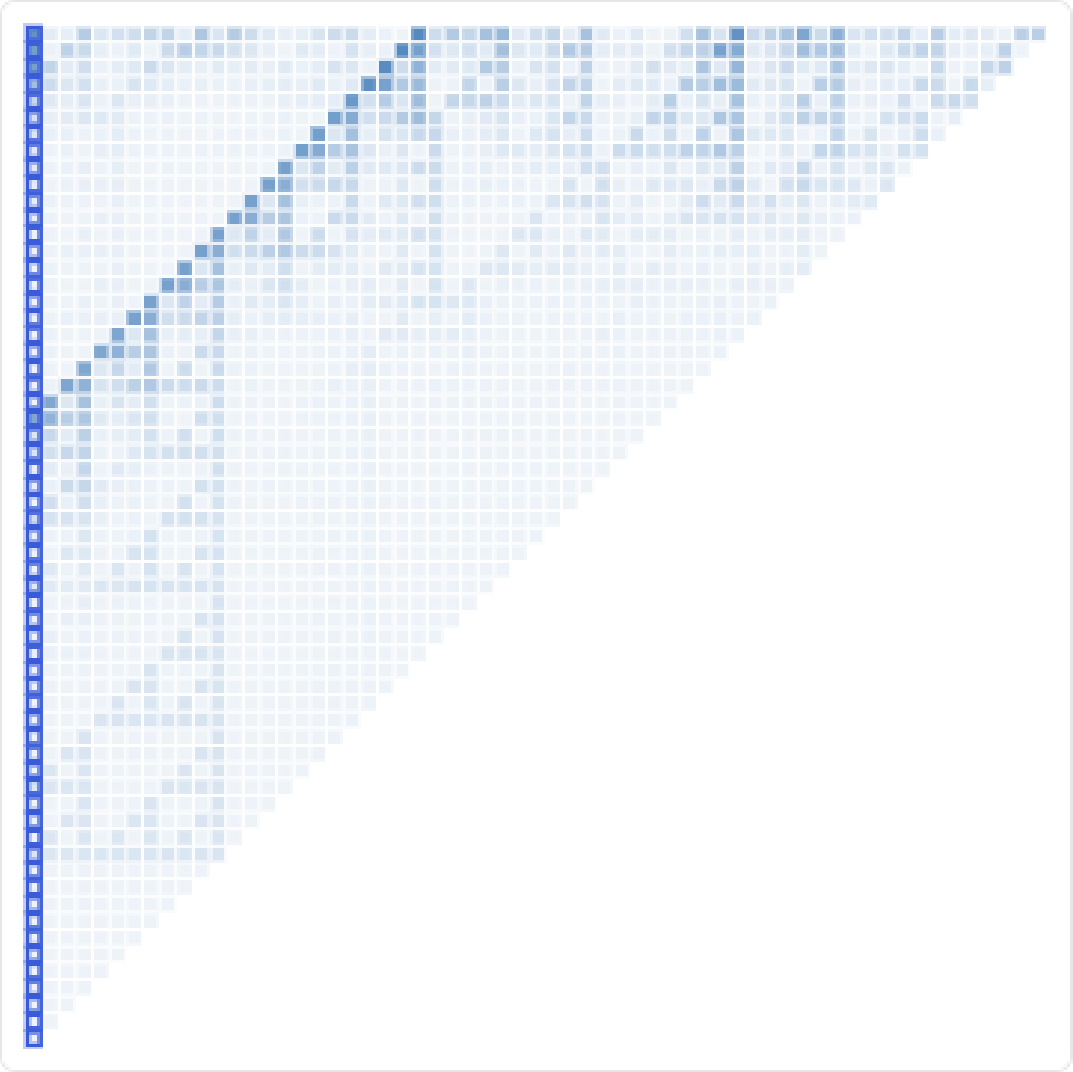


Here is what an $n = 64$ version of this picture looks like (with the spike positioned at the 32th entry):



The number of 1s in the kth row is then $2^{s_2(k)}$ (if we index the rows starting from zero), which is at least of the same shape as the empirical prediction, albeit with different constants. (This sequence is also known as Gould's sequence.)

Numerically, we seem to observe fragments of Sierpinski gaskets being generated before decaying (often due to "collisions" with other gaskets):



However, it is not clear to me at all what the asymptotic probabilistic model should be, even heuristically; it does not resemble any random shape model that I am familiar with. But perhaps there are readers more expert in probability theory or statistical physics who may be able to suggest such an asymptotic limit?

Comments (6)

Steven Heilman · 14 July 2026

Rule 90 (one-dimensional Conway's Game of Life) also exhibits Sierpinski triangles, though the connection to Gilbreath's conjecture is known, so you all probably knew this already.

Anonymous · 16 July 2026

And here are some references on that simpler case:
M. Miyamoto, "An equilibrium state for a one-dimensional life game," *Journal of Mathematics of Kyoto University***19**(3) (1979), 525–540.
Douglas A. Lind, "Applications of ergodic theory and sofic systems to cellular automata," *Physica D***10**(1–2) (1984), 36–44.
M. Miyamoto, "Stationary measures for automaton rules 90 and 150," *Journal of Mathematics of Kyoto University* **34** (1994), 531–538.
M. Takei, "Limiting measure for addition modulo a prime number cellular automata," *International Journal of Networking and Computing* **7** (2017), 124–135

Anonymous · 15 July 2026

Sorry if offtopic, but what's your agent workflow? Standard Claude Code, or something more refined? Is there any agent you find is best suited for this type of work?

> 
> **Terence Tao** · 15 July 2026
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> I am using a standard Claude Code agent (running Opus 4.8) that maintains a skill file for my github repository that records all the lessons learned through the various applet projects I have already conducted, and is aware of my preferences and design philosophy. But I imagine that many other modern AI agents would be capable of similar assistance in generating applets of this type; as I stated in a previous post, the raw coding power of the agent is no longer the primary bottleneck, with most of the non-automated work now being concentrated in design, specification, and review.

garylangford · 17 July 2026

Is there a more fundamental operator or state-space whose invariant measure produces the expectation sequence cnc_ncn, with the observed Sierpinski fragments emerging as transient structures of that dynamics rather than as the primary object?

Anonymous · 24 July 2026

The graph of c_n by n is visually interesting! Looking a little past the bumpiness of the graph (where c_n is smaller for even n than for neighboring odd values), c_n seems to drop especially precipitously for n that are powers of 2.
Aside:
(I used the visualizer app you linked to sweep a spike across different starting positions for sizes close to a power of two. The view is similar to a sliding window over a Sierpinski triangle (though there may be some "edge effects" that I will ignore for now).
Let's look at the density of "cell occupancy" for a row in the Sierpinski triangle at different heights. At heights that are a power of 2, all cells in a consecutive run are occupied (there are horizontal lines). Just above that at -1 + 2^k you see the cells are about 50% occupied, and just below that, at 1 + 2^k, there is mass "cancellation" where almost no cells are occupied. For a nonzero value to penetrate all the way to the bottom, the spike must start at one of the endpoints of the top row.
This made me wonder if 1. could there be an off-by-one discrepancy between the sizes of the app and the values in the graph, since in the app **1 +** 2^k sizes are especially unlikely to allow a spike to propagate to the bottom, while in the graph, it is when n = **2^k** itself that c has a low expected value; or perhaps 2. the behavior of spikes is just too different from when exponential random variables are used?)